INCH-POUND

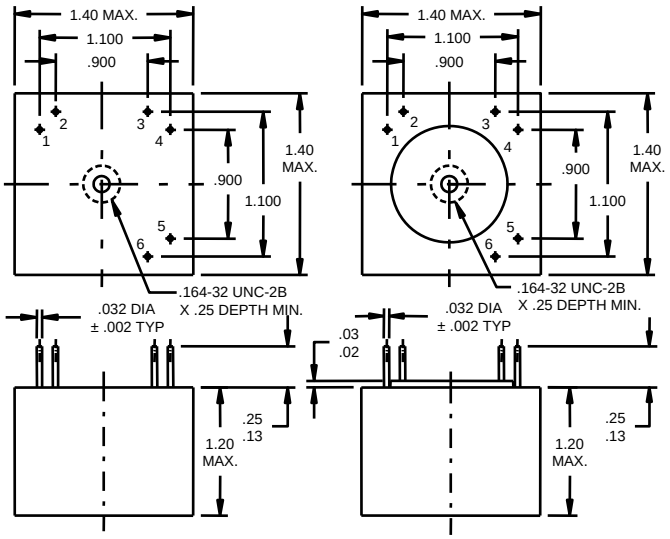
MIL-PRF-27/46C
29 September 2009
SUPERSEDING
MIL-PRF-27/46B
7 February 1984

PERFORMANCE SPECIFICATION SHEET

# TRANSFORMERS, POWER, 25 VOLT AMPERES, 400 HZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and MIL-PRF-27.




| Inches | mm |
|---|---|
| .002 | 0.05 |
| .02 | 0.5 |
| .03 | 0.8 |
| .032 | 0.81 |
| .13 | 3.3 |
| .164 | 4.17 |
| .25 | 6.4 |
| .900 | 22.86 |
| 1.00 | 25.4 |
| 1.10 | 27.9 |
| 1.20 | 30.5 |
| 1.40 | 35.6 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations



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REQUIREMENTS: (When numbers in parentheses, such as (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 4.4 ounces, maximum.

Duty cycle: Continuous.

Case: Plastic encapsulated.

Material: Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55˚C to +130˚C.

Terminals: Printed circuit pins – tinned brass.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts rms, 400 Hz across (1-2): Current in (1-2) shall not exceed 25 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a restrictive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 88 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no Load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

shall not exceed 7 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 45°C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.



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Marking location: See figure 1.

VERIFICATION

Extent of qualification

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-PRF-27/37 through MIL-PRF-27/47 inclusive, all parts.

Qualification testing and approval to M27/46-44 shall be sufficient to grant qualification approval to M27/46-01 through M27/46-44.

Part or Identifying Number (PIN), example:

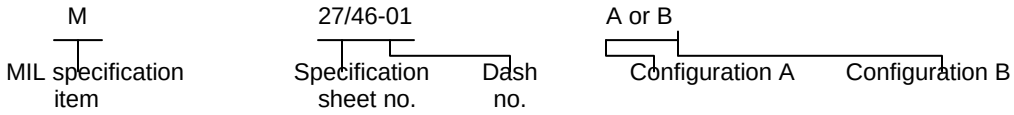




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TABLE I. Electrical ratings.

| Dash No. | Type designation | Secondary voltage (3-5) (V rms) ±3% | Secondary current (3-5) (amps rms) | Working voltage (V rms) (MAX) |
|---|---|---|---|---|
| 01 A or B | 796 | 6 | 4.167 | 325 |
| 02 A or B | 797 | 6.3 | 3.968 | 325 |
| 03 A or B | 798 | 7 | 3.571 | 325 |
| 04 A or B | 799 | 7.5 | 3.333 | 325 |
| 05 A or B | 800 | 8 | 3.125 | 325 |
| 06 A or B | 801 | 9 | 2.778 | 325 |
| 07 A or B | 802 | 10 | 2.500 | 325 |
| 08 A or B | 803 | 11 | 2.273 | 325 |
| 09 A or B | 804 | 12 | 2.083 | 325 |
| 10 A or B | 805 | 12.6 | 1.984 | 325 |
| 11 A or B | 806 | 13 | 1.923 | 325 |
| 12 A or B | 807 | 15 | 1.667 | 325 |
| 13 A or B | 808 | 16 | 1.563 | 325 |
| 14 A or B | 809 | 18 | 1.389 | 325 |
| 15 A or B | 810 | 20 | 1.250 | 325 |
| 16 A or B | 811 | 22 | 1.136 | 325 |
| 17 A or B | 812 | 24 | 1.042 | 325 |
| 18 A or B | 813 | 26 | 0.962 | 325 |
| 19 A or B | 814 | 28 | 0.893 | 325 |
| 20 A or B | 815 | 30 | 0.833 | 325 |
| 21 A or B | 816 | 33 | 0.758 | 325 |
| 22 A or B | 817 | 36 | 0.694 | 325 |
| 23 A or B | 818 | 39 | 0.641 | 325 |
| 24 A or B | 819 | 43 | 0.581 | 325 |
| 25 A or B | 820 | 47 | 0.532 | 325 |
| 26 A or B | 821 | 51 | 0.490 | 325 |
| 27 A or B | 822 | 56 | 0.446 | 325 |
| 28 A or B | 823 | 62 | 0.403 | 325 |
| 29 A or B | 824 | 68 | 0.367 | 325 |
| 30 A or B | 825 | 75 | 0.333 | 325 |
| 31 A or B | 826 | 82 | 0.305 | 325 |
| 32 A or B | 827 | 91 | 0.275 | 325 |
| 33 A or B | 828 | 100 | 0.250 | 325 |
| 34 A or B | 829 | 110 | 0.227 | 325 |
| 35 A or B | 830 | 115 | 0.218 | 325 |
| 36 A or B | 831 | 120 | 0.208 | 325 |
| 37 A or B | 832 | 130 | 0.192 | 325 |
| 38 A or B | 833 | 150 | 0.166 | 325 |
| 39 A or B | 834 | 180 | 0.139 | 325 |
| 40 A or B | 835 | 200 | 0.125 | 440 |
| 41 A or B | 836 | 220 | 0.113 | 440 |
| 42 A or B | 837 | 240 | 0.104 | 440 |
| 43 A or B | 838 | 270 | 0.093 | 580 |
| 44 A or B | 839 | 300 | 0.083 | 580 |



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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

| | | | |
|---|---|---|---|
| MIL-PRF-27/37 | MIL-PRF-27/38 | MIL-PRF-27/39 | MIL-PRF-27/40 |
| MIL-PRF-27/41 | MIL-PRF-27/42 | MIL-PRF-27/43 | MIL-PRF-27/44 |
| MIL-PRF-27/45 | MIL-PRF-27/47 | MIL-STD-202 | |

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2009-024)

Review activities:
Army - CR4, MI
Navy - AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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